My white board writing from week 6

1) I went through Lab3 first. It discusses working with SAS libraries, importing different types of data into SAS and creating SAS data files (with extension .sas7bdat). I also discussed and demonstrated how you can create your own SAS library and how do you refer to the SAS data files contained in this library. The related handout is thorough enough and I do not reproduce it again here. You are advised to experiment in the lab with these commands in your leisure time.

2) Then I went through lecture 6: Discrimination and Classification. The slides / notes are detailed enough and I am not repeating the derivations here again. I just want to stress on the graphical illustration of the decisions based on the Fisher discriminant function.

Assume that we are trying to discriminate between two multivariate normal populations with densities: $f_1(x)$ being $N_p(\mu_1, \Sigma_1)$ and $f_2(x)$ being $N_p(\mu_2, \Sigma_2)$ and assume that $\Sigma_1 = \Sigma_2 = \Sigma$. Also, assume that the priors p_1, and $p_2 = 1 - p_1$ are the same $(p_1 = p_2 = \frac{1}{2})$ and the costs of misclassification are also the same. Then the general inequality defining the region R_1 (which originally is given by

$$R_1 = \{x : \exp[-\tfrac{1}{2}(x-\mu_1)'\Sigma^{-1}(x-\mu_1) + \tfrac{1}{2}(x-\mu_2)'\Sigma^{-1}(x-\mu_2)] \geq \frac{c(1|2)}{c(2|1)} \frac{p_1}{p_2}\})$$

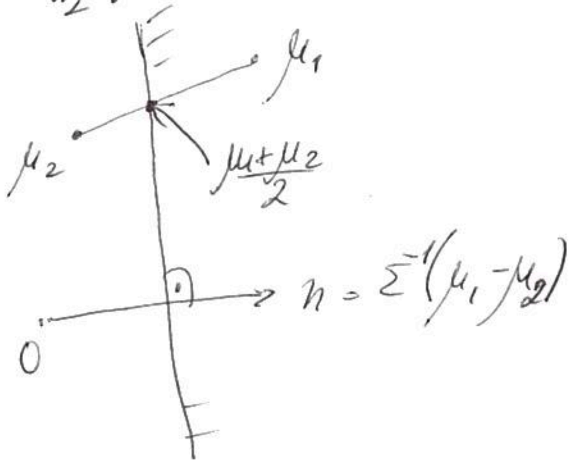
reduces to

$$R_1 = \left\{ X : (\mu_1-\mu_2)'\Sigma^{-1}X - \frac{1}{2}(\mu_1-\mu_2)'\Sigma^{-1}(\mu_1+\mu_2) \geq 0 \right\}$$

(since $\log\left[\frac{c(1|2)}{c(2|1)} * \frac{p_2}{p_1}\right] = 0$).

Denote by n the p-dim vector $n = \Sigma^{-1}(\mu_1-\mu_2)$ and let X_0 be (any) new observation vector that we want to classify into population Π_1 or Π_2.

Consider the (line for $p=2$ or more generally the hyperplane for $p>2$) that has n as a normal vector and that passes through the point $\frac{\mu_1+\mu_2}{2}$.



For any X_0 on that hyperplane, $X_0 - \frac{1}{2}(\mu_1+\mu_2)$ belongs to the hyperplane, hence $n^T\left(X_0 - \frac{1}{2}(\mu_1+\mu_2)\right) = 0$ holds.

If X_0 is on the side of the plane then the vector $X_0 - \frac{1}{2}(\mu_1+\mu_2)$ and the vector n "cross" at an angle wich is <u>acute</u> hence their scalar product is <u>positive</u>. Vice versa, if X_0 is on the side of the plane then the vector $X_0 - \frac{1}{2}(\mu_1+\mu_2)$ and the vector n "cross" at an angle which is <u>obtuse</u> and as a result their scalar product is <u>negative</u>.

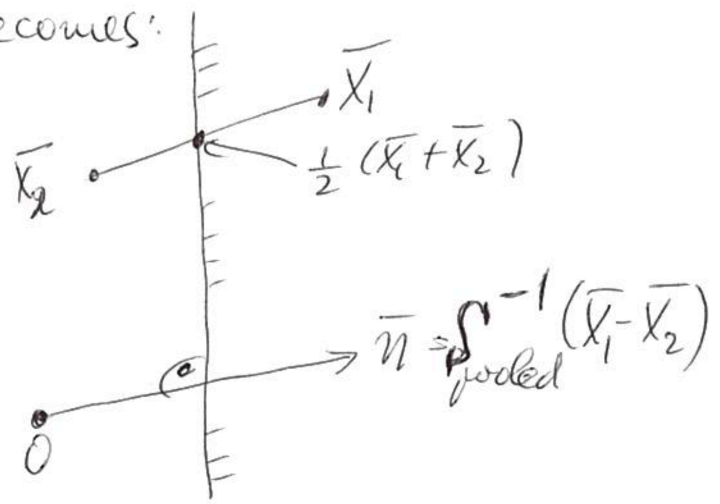
In the first case we classify into Π_1 and in the second case — into Π_2. The decision is <u>simple</u>: look whether

the point to be classified is above or below the hyperplane.
The above discussion was "at a population level". In an inferential context, we will have the __learning sample__: some observations $X_{11}, X_{12}, \dots, X_{1n_1}$ for which it is __known__ that they belong to π_1 and some other observations $X_{21}, X_{22}, \dots, X_{2n_2}$ for which it is known that they belong to π_2.
Then we can estimate $\hat{\mu}_1 = \bar{X}_1$, $\hat{\mu}_2 = \bar{X}_2$ by using the component-wise averages and estimate the joint Σ by the pooled $S_{pooled} = \frac{1}{n_1+n_2-2}\left[(n_1-1)S_1 + (n_2-1)S_2\right]$

with $S_1 = \frac{1}{n_1-1}\sum_{i=1}^{n_1}(X_{1i}-\bar{X}_1)(X_{1i}-\bar{X}_1)'$; $S_2 = \frac{1}{n_2-1}\sum_{i=1}^{n_2}(X_{2i}-\bar{X}_2)(X_{2i}-\bar{X}_2)'$

The previous graph now becomes:

and we proceed with the discrimination in a similar way as before



3) I then discussed the case of more than 2 populations where the discriminant score for each of the $g>2$ populations needs to be calculated in order to classify into the population with the largest discriminant score. This is discussed thoroughly in the lecture notes.